RUIPU LI

Third-year Ph.D. Student in Computer Science and Engineering, University of Michigan

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Research interests: uncertainty quantification for time series forecasting with conformal prediction, flow matching and Bayesian inference; agentic frameworks; real-time time series forecasting; epidemic scenario modeling

EDUCATION

Ph.D., Computer Science and Engineering, University of Michigan, Ann Arbor *Aug 2024 – Present*
Advisor: Alexander Rodríguez, Complexity AI Lab. Qualifying and preliminary exams passed in the first year.

M.S.E., Computer Science and Engineering, University of Michigan, Ann Arbor *May 2024*

B.S.E., Computer Science, University of Michigan, Ann Arbor *Apr 2022*
UM–SJTU Joint Institute dual-degree program.

B.S., Electrical and Computer Engineering, Shanghai Jiao Tong University *2018 – 2022*

PUBLICATIONS

Li, R. & Rodríguez, A. *Neural Conformal Control for Time Series Forecasting.*
39th AAAI Conference on Artificial Intelligence (AAAI-25), main track, 2025 · arXiv:2412.18144 · [Code]

Li, R., Menacho, D. & Rodríguez, A. *Optimization-based Online Conformal Prediction for Multi-step Forecasting.*
The 5th Workshop on Uncertainty Reasoning and Quantification in Decision Making (UDM-KDD'26), **accepted as oral** · under review, AAAI-26 · arXiv:2508.13362

Yang, Y., **Li, R.** & Rodríguez, A. *Simulate, Reason, Decide: Scientific Reasoning with LLMs for Simulation-Driven Decision Making.*
SciSoc Agents & LLMs Workshop, KDD 2026, **accepted as oral** · under review, NeurIPS 2026 · arXiv:2606.04505

RESEARCH EXPERIENCE

FluSight Forecasting Challenge – Centers of Disease Control and Prevention (CDC) *Jul 2023 – Present*
Team Lead, UM-DeepOutbreak, University of Michigan

- Leads weekly submissions for three CDC forecast hubs. Achieved **top-3 accuracy** (MAE) out of ~40 teams in the 2023–24 FluSight forecast challenge.
- Built and maintain the team's end-to-end forecasting pipeline, encompassing data ingestion, deep learning model training, ensembling, uncertainty quantification, and evaluation.
- Developed conformal prediction methods for reliable uncertainty quantification in real-time, multi-step forecasting, leading to Neural Conformal Control (NCC) and Optimization-based Online Conformal Prediction (O^2CP).

Infectious Disease Modeling *Feb 2026 – Present*
Collaboration with the Michigan Public Health Integrated Center for Outbreak Analytics and Modeling (MICOM)

- **ScenarioPilot** (since May 2026): Collaborate with epidemiologists to develop an agentic framework that automates infectious disease modeling, enabling faster and more reliable exploration of scenario assumptions.
- **Gray-box scenario modeling with flow matching** (since Jun 2026): Develop a gray-box framework that couples a mechanistic epidemic model with flow matching to jointly learn the latent epidemic parameters and the observation process, combining data-driven learning with domain-knowledge-informed priors to generate probabilistic scenario projections.

Uncertainty Quantification for Cyclone Trajectory Prediction *Oct 2025 – Present*
Collaboration with Google DeepMind WeatherLab

- Apply our adaptive conformal method (O^2CP) to ensemble cyclone forecasts to construct multi-step prediction sets with long-term coverage guarantees.
- Compared with the raw prediction sets from Google WeatherLab, O^2CP reaches coverage close to the nominal 95% while the existing cones under-cover at several horizons.

Machine Learning to Reduce Uncertainty in Aerosol Climate Forcing *Aug 2022 – Aug 2024*
Research Assistant, Climate and Space Sciences and Engineering, U-M (Advisor: Joyce Penner)

- Developed machine learning models from satellite observations to predict aerosol optical depth and cloud droplet number concentration, using PCA, recursive feature elimination, and SHAP to identify key predictive drivers.

INDUSTRY EXPERIENCE

NIO – Machine Learning Intern, Autonomous Driving System *May 2021 – Aug 2021*

- Improved autonomous lane-change decisions with an LSTM model that predicts surrounding vehicles' motion; curated large-scale road-test logs into a modeling dataset and built Python analysis tooling with Google Test.

HONORS AND AWARDS

Graduated *summa cum laude*, University of Michigan	2022
University Honors (three terms), University of Michigan	2021 – 2022
Dean's List, every semester in residence, University of Michigan	2020 – 2022
Undergraduate Excellent Scholarship, Shanghai Jiao Tong University	2019
John Wu and Jane Sun Excellence Scholarship (top 3%), Shanghai Jiao Tong University	2018

INVITED TALKS

"Adaptive Conformal Prediction in Real-World Time Series Forecasting," MIDAS Summer Academy	2026
"LLMs for Epidemic Scenario Modeling," MICOM Seminar Series (MDHHS and public health partners)	2026
"Neural Conformal Control for Time Series Forecasting," Applied Conformal Prediction course (Maven)	2025

TEACHING AND MENTORING

Graduate Student Instructor, EECS 492: Introduction to Artificial Intelligence *Winter 2026, Fall 2026*

- Gave lectures on intro machine learning and decision trees, wrote the weekly homework, and ran discussion sections, office hours, and grading.

Undergraduate Research Mentor (UROP), University of Michigan *2025 – 2026*

- Mentored an undergraduate working on the CDC flu forecasting pipeline; she presented at the UROP symposium and stayed on in the lab as a volunteer.

SERVICE

Program Committee, epiDAMIK Workshop @ KDD	2024, 2026
Reviewer, NeurIPS Workshop on Foundation Models for Temporal Systems (FMTS)	2026

TECHNICAL SKILLS

Methods: conformal prediction (online and adaptive), probabilistic multi-step forecasting, flow matching and diffusion models, Bayesian inference and particle filtering, compartmental epidemic models (SIR/SEIR).

Tools: Python, PyTorch, R, C/C++, SQL; LangGraph, LangChain; Slurm/HPC, GCP, BigQuery.